Spring 2024 – User Research Methods – Professor Charles Raffaele
Video Observation Design Document – Darryl Hutchinson

A. INTRODUCTION

Though they have great potential for entertainment and learning, video games are typically very complex and it can be easy for inexperienced players to get overwhelmed and lose site of the objectives the designer wants them to experience. To better understand intuitive level design and which game elements do the best job of capturing player attention regardless of experience level such that they don't get lost and can progress through an experience, I want to analyze the upcoming video game, Europa.

Designed and developed by Helder Pinto and Novadust studios, Europa is a bright and colorful Sci-Fi adventure game with a heavy influence of painterly Ghibli style visual animation, smooth freeform movement and traversal with low stress exploration. The game has been in development since 2017 and is currently available in demo form with a scheduled release window of spring 2024.

I am interested in exploring pacing/pathing and environmental storytelling. I would like to know specifically how game elements in the game world can communicate which ways players can travel without using text or unnatural UI markers. Since video games are relatively complex bundles of systems that are easy to get lost within, I want to try and understand how to better lead the player along an experience with natural elements that help maintain immersion.

The target audience for this research could be the developers Novadust studios themselves to gain more perspective on what their players are experiencing. With more feedback about what is drawing player attention, adjustments can be made to the level design to further avoid players getting lost or confused in the game. Another audience could also be other developers, researchers and/or game design enthusiasts who want to more deeply understand how specific elements in Europa create specific responses or perceptions in players.

This analysis could easily be applied to another game, but I am choosing the same game as the previous think-aloud protocol analysis in order to keep as many variables as consistent as possible to more clearly compare/contrast the differences in observation approach. Highlighting these differences will make generalizable patterns more easily identifiable and potentially applicable to future projects.

The results of this video observation analysis will be submitted to Professor Raffaele for evaluation.

B. VIDEO OBSERVATION RESEARCH QUESTIONS:

1. **What objects/colors/landmarks attract the most player attention?**
    a. How much time is spent at each object and/or major landmark
        i. Coded* for by: time spent (TS), Success (S / U), questions (QA)
    b. What are players trying to do at each attention center?
2. **How do players react to various in game items/prompts/abilities/obstacles?**
    a. Which ones are clear/unclear for the participant?
        i. Coded* for by: utterance/expression (RVU / RFE), time spent (TS), questions (QA), prompt reaction (PR / PI)

3. **What actions does the player attempt that the game does not allow?**
    a. What confuses the player?
        i. Coded* for by: Utterance(RVU / RFE), Success (S / U), Questions Asked (QA), task accuracy (T / O)

Please find a coding explanation below in section D. Codes and Construct Variables
****Please find a sample coding scheme below in figure 1***

C. Study Design

Video Observation will be used this time in order to try clearly capture participant behavior and retain a record of said behavior for convenient future reference. Rather than standard observation protocols and think-aloud protocols where every behavior can be difficult to record and cannot be accessed again to confirm accuracy, video observation will allow us to keep a detailed record of everything the participant does in the game. Video observation is also less intrusive and the participant is expected to feel more at ease and relaxed compared to think-aloud protocols.

Conducting video observation does come at somewhat of a cost in that analyzing this recording will take much longer than what might be typically required for standard observation and/or think-aloud analysis. However, we have a crack team of experienced research professionals ready to process this data.

Having a documented and easily re-accessible account of everything researchers detect will make time more time, but ultimately allow for easier future reference. Having detailed and accurate coding from researchers with clear visual observation data from participants will give developers an extremely high resolution look at how participants are interacting with their software.

The participant will be set-up to play through the demo, Europa. They will be given a simple set of tasks and their behaviors as they proceed through said tasks, will be recorded. There will be one recording of gameplay and another recording of the participant's face.

Testing has not yet completed, but the researcher will be using one of the following video recording software options: Camtasia, ScreenFlow, OBS Studio, Datavyu.

Participant Personas

This participant is a personal relationship. They have little to no experience with video game culture, is 70 years old, identifies as male and his highest level of formal education is college. The participant is also now retired. The participant claims not to play video games on any regular basis and has little to no familiarity with modern game design.

Given enough time, I would also like to conduct this experiment with another relatively gaming-inexperienced participant from a different age demographic to compare and contrast behavior. The target participant would be in their 20s or 30s, identify as female and highest level of education would be university. The participant would likely be working a day job and claim not to play video games on any regular basis nor has any familiarity with modern game design.

D. VIDEO OBSERVATION INSTRUMENTS & PROCEDURE

Technical Set-up

1. Prepare the test setup hardware (table, chair, laptop PC with keyboard, trackpad and external wireless mouse, and USB PlayStation 4 controller)
2. Prepare the test setup software (Steam, Europa Demo)
    a. Check controller inputs, video resolution, frame rate, etc. to make sure game is running properly
3. Prepare the video recording hardware (PC, Microphone/Webcam)
4. Prepare the video recording software (Camtasia, ScreenFlow, OBS Studio, Datavyu)
    a. Testing will continue up until session date before selection is made
5. Prepare coding scheme spreadsheet
6. Confirm recording functionality with a 10-second test recording
    a. Make sure both audio and video properly playback

Procedural Set-Up

1. Prepare warm up questions to welcome and relax the participant
2. Participant will be reminded about research goals and rights to privacy/confidentiality
3. Participant will be reminded to relax since there is no performance based evaluation
4. Once technical aspects have been confirmed to be in working order and participant is comfortable, researcher can start recording and commence the session

Tasks

Because thinking aloud while playing a game isn't necessarily standard behavior for everyone, and the participant may forget to keep talking, some task-specific probes will be provided here for quick reference:

1. Do you understand the objective?
2. Please collect the emeralds in this area.
3. Please find the messages in this area.
4. Please obtain the abilities in this area.
5. Please proceed to the next area.

*Codes and Construct Variables

- TS - **T**ime **S**pent
    - (duration of **t**ime **s**pent at point of interest) RQ2
- RVU / RFE – **U**tterances / **E**xpressions
    - (**R**eactions **V**erbal **U**tterances/ **R**eactions **F**acial **E**xpression) RQ
- S / U – **S**uccessful / **u**nsuccessful
    - (**S**uccessfully/**u**nsuccessfully understood objective and could/couldn't clear obstacle)
- T / O – On **T**ask or **O**ff Task
    - (Is participant on **t**ask and making progressing towards next objective or **o**ff task meandering)

- QA – **Q**uestions **a**sked
    - (# of **q**uestions **a**sked to researcher for help or clarification)
    - (content of **q**uestion **a**sked for assistance)
- PR / PI – In-game **P**rompt **R**ecognized or **P**rompt **I**gnored
    - When an in-game prompt or message appears, does the participant recognize and react accordingly or ignore the prompt.

Sample Coding Scheme Spreadsheet

	A	B	C	D	E	F	G	H	I	J
1	CONSTRUCT		MEASURE		OBSERVABLE BEHAVIOR / INDICATOR		CODE			
2										
3	Time Spent		How much time participant spends @POI		Stopping at a POI		**TS**	**T**ime **S**pent		
4					Stopping anywhere and spending time					
5										
6	Utterances/Expressions		What does particpant say re: stimulus?		Making remarks		**RVU / RFE**	**R**eaction **V**erbal **U**tterance /		
7			What expression does participant make?		Making facial expressions			**R**eaction **F**acial **E**xpression		
8										
9	Successful/Unsuccessful Task		Can participant complete tasks?		Can partipant explain objective?		**S / U**	**S**uccessful/**U**nsuccessful		
10					How many emeralds found?					
11										
12	On or Off Task		Is participant proceeding to task?		Is participants proceeding towards next POI?		**T / O**	On-**T**ask / **O**ff-Task		
13			Or is partiants meandering away from task?		Is participant moving away from next POI?					
14										
15	Questions Asked		What questions does the participant want answered?		# of questions participant asks		**QA**	**Q**uestions **A**sked		
16					Theme or pattern of questions asked					
17										
18	Prompt Reactions		How does participant respond to game prompts?		Does participant do anything in reaction to prompt?		**PR / PI**	**P**rompt **R**ecognized /		
19					What does participant say/do after reading prompt?			**P**rompt **I**gnored		

Figure 1, sample coding scheme

E. DATA ANALYSIS PLAN

After the session is complete the video data will be saved and backed up to another drive for emergency backup. A minimum crack team of at least two researchers will then simultaneously commence analysis by reviewing video data while coding for target constructs. Upon completion of video review and construct coding, each researcher will cross-reference their results to solidify researcher inter-reliability. Researchers A and B can also compare notes to expose any inconsistencies and to confirm any potential patterns of interest and/or correlations in observed participant behavior.

These results will be then be written up into a brief report that includes the participant's profile, the coding scheme used, the instruments used, the observation procedure employed as well as any utterances, reactions or other video excerpts that demonstrate any particularly interesting correlations. Upon request, a brief slide show presentation consolidating all of this data can additionally be prepared and delivered.